

Solutions to Encrypted Mess

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Encrypted password: **ETHAN_SHARMA_THE_GOAT**

Solution. It would be great if we first figured out what we needed to do. The “story,” if one can call it that, provides hints as to the steps needing to be taken to change this sequence of numbers into a string of characters. The first hint is in the phrase “A skii.” Reading this fast, this should give a somewhat similar word to “ASCII,” which is exactly what we’re looking for. So, working backwards, we know that our last step is going to be converting from ASCII into characters.

The next hint involves the 3-fingered cat. An apt mathematical synonym for finger is *digit*, and cat is part of the word “concatenate.” Perhaps a bit of a stretch, but there isn’t much else that would make sense. So, our next step would be combining 3 numbers together at a time (hence, 3 ASCII characters). The *firsts*, which can be interpreted as *prime*, of July or November, which are the 7th and 11th months respectively, meaning that each of these concatenations must be multiplied randomly by either 7 or 11. Some random constants are then added to each concatenation, and after the remaining digits are concatenated, the whole number is squared. This number then gets the sum of the previous random numbers added on to its end.

Working backwards, we should find the greatest square less than this number. Through great experimentation and trouble, it is possible to find that our number $K - 18$ is equal to 76831937236848586085791054191054319876952825576089 squared. Perhaps a calculator would be nice.

Through some experimentation, after dividing $\sqrt{K - 18}$ into units of 6 digits each, we can find the original sequence in which numbers have randomly been multiplied by 7 or 11.

7683192 7236845 5860855 9105415 10543192 7695281 5576088

Some of these are either 7 or 8 digits. This is because the original sequence is comprised of 6 digits for each of these, and multiplying a sufficiently large 6 digit number by 11 can result in an 8 digit number. Dividing wisely by either 7 or 11, we get

698472 657895 837265 827765 958472 699571 796584

and separating these, we get sequences of two digit ASCII characters, which when converted, give the desired result.